

Exploration with *Mathematica*

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During the Fall Semester of 1989 I presented a junior/senior mathematics course on **projective geometry** at my university, with a syllabus organized along quite traditional lines. I used a Macintosh II computer and the *Mathematica* symbolic computation program, a product of Wolfram Research, Inc. The lectures were delivered primarily from the console, using a screen attached to the computer and projected by an overhead projector. The students were expected to do their homework on a computer, and the final examination was a project on a topic selected individually by each student, again, to be done on the computer.

During the present Fall Semester of 1990 I am presenting a course on **methods of symbolic computation**. The audience is very mixed, and we are doing a selection of topics, some selected by the students and some selected by the instructor. An attempt is being made to tie in the individual work of the students with work for other courses. This year we have a classroom with a cluster of NeXT computers, with 12 for the students and one for the lecturer, and each machine runs *Mathematica*. There is a file server for distributing the class materials. A better projection device is being used this year giving a large image of what the instructor has on his screen. The cluster is open 24 hours a day, but it is reserved six hours a week for class work in two different classes. This is an experimental installation for testing teaching methods. Evaluation of student work again will be done by individual projects.

In my presentation, I will demonstrate on a computer several examples from these courses, and show how I use the machine in teaching. I will then discuss several lessons gained from the experience and suggestions I have for future developments.